

FINANCIAL LIBERALIZATION, ECONOMIC FREEDOM AND THE BANKING CRISIS

Dr. Anjala Kalsie* Jappanjyot Kaur Kalra**

Abstract:

One of the most important policy issues being discussed in India since the 1991 crises is whether to liberalize the Capital Account completely or not. In this paper we make an attempt to analyze the empirical relationship between the Financial Liberalization and the Banking Crises with an impact of Economic Freedom. The data set adopted for the purpose of study is for the years 1980-2005 and covering Emerging Markets: BRICS (Brazil, Russia, India, China and South Africa), ASEAN 5 [Indonesia, Korea, Malaysia, Philippines and Thailand (those impacted by 1997-98 crises)]. The various previous studies have argued that liberalization if accompanied by weak prudential supervision and insufficient financial reforms augments the likelihood of the banking crises. This papers result supports the cautious approach towards liberalization with weak institutional environment and impact over the various dimensions of the financial reforms using the panel Probit model.

Keywords: Financial Liberalization, Banking Crises, Economic Freedom, Institutional Environment, Probit Model.

JEL Classification: E44; G21; G28; F36

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1. Introduction

Since many decades the researchers and the policy makers are discussing the relationship between the financial liberalization and financial crisis but no definite conclusion has been derived. The critiques to financial liberalization like “Minsky” who gave the financial instability hypothesis(Minsky, 1993) concludes that in the long run the financial institutions and the financial markets are not stable, financial liberalization shows positive expectations in the short run only. The liberalization leads to higher flow of capital into the markets which are speculative in nature and these flows are invested into the riskier projects as to earn higher returns. These situations only show positive and greater impact on the economic growth in the initial years of capital account openness and later on as the risks and the defaults increases reversing the situation and leading to instability and crisis in various sectors the economy mainly the banking sector. The experiences of the various Latin American nations and the South East Asian nations support the fact that the liberalization policies till date are not efficient and there are no common views on it by the researchers(B. Eichengreen, 2001). He in his paper mentioned that removing controls on the capital account may not be beneficial in the presence of the other distortions which are plausibly the case in the developing economies.

There are lot of empirical studies done on the link between the financial development and economic growth,(Edison, Levine, Ricci, & Sløk, 2002; Levine, 2001) who support the liberalization of the financial sector as it leads to development by stimulating economic growth through channelizing the savings and other economic resources following the McKinnon and Shaw(1973) theory of financial repression and financial development.

Various studies (Beck, Demirgüç-Kunt, & Levine, 2006; A Demirgüç-Kunt & Detragiache, 1998; Mehrez & Kaufmann, 2000; Shehzad & Haan, 2009) focuses on the issue that more open economies are prone to crisis in particular the banking sector¹ becomes more fragile due to the frequent inflows and outflows of the funds. The capital account convertibility (CAC) induces the behaviour of higher risk taking by investing into riskier projects in expectation of higher returns from the emerging markets. But these studies basically used very objective data for liberalization i.e. just focusing on the deregulation of the banking interest rates. It is a cycle where investors invest in the fast growing economies where the opportunity cost is higher, (the herd behaviour of the investor follows), thus reducing the cost of capital (pooling liquidity risk (Diamond & Dybvig, 1983)) and the returns on investment. Thus the investors start moving out the money from these projects in emerging markets again the herd behaviour of investors follow, the sudden inflows and outflows in the economy traumatize the banking sector. The openness and the removal of the controls on the international capital flows render the financial intermediaries to jeopardy of foreign exchange risk. (Graciela L. Kaminsky & Reinhart, 1999) states that the foreign currency risk is transformed to the credit risk (the financial intermediaries raise funds in foreign currency abroad and lend it to domestic borrowers) causing the twin crises (both currency and banking crisis).

The paper is an attempt to analyze empirically the theoretical reasons that the capital account open economies, in particular the liberalized banking system makes economy more vulnerable to crisis. The primary function of the banking system is to channelize the short-term deposits into the long term funds (loans) matching the maturities. The theory states that in the more closed economies the bank lending interest rates are subject to ceilings, due to which they can't invest or lend loans in the high risk projects and makes it impossible for these banks to charge the high risk premium and earn higher returns. In an open economy the interest rates (nominal) are market determined and are more volatile² as more variables are also associated. Once the financial system is deregulated i.e. the ceilings are lifted, more investing participants enter into the competition i.e. it becomes easier for the banks to lend and disburse the funds to the more risky projects, charging high risk premia and earning

¹ “Chilean experiences, and those of other Latin American countries, also show that fiscal extravagance is a sure way to bring about not only economic dislocation, but also the weakening and even collapse of fragile democratic institutions” (Diaz-Alejandro, 1985). The same situation was faced by the Asian tigers or the fast growing Asian economies which suffered with crisis in 1997-98 after the fall of Thai Bhatt.

² “Bank insolvency often occurs when macroeconomic volatility is on the rise and when incentives appear to have weakened— not a good time to try to determine causality. When financial deregulation occurs at the same time it becomes a convenient target, though in many countries deregulation proceeded faster than improvements in financial infrastructure and incentives, suggesting that weak financial infrastructure and poor incentives led to the crises” (Caprio & Klingebiel, 1997).

higher returns. These investment bankers hedge the credit risk by holding well-diversified portfolios, but these portfolios are exposed to the various risk and economy wide shocks. In a less developed banking system the maturity mismatch can occur leading to fund shortages in the system leading to liquidity problem and contagion due to market imperfections and panic among the public. The evaluation and the monitoring of these loans and the collateral is a primary activity and require lot of skills, which may not be the case with all of these banks. Thus the failure to evaluate the loan during the term (along with other factors like economic slowdown, reduction in demand etc) leads to the non payment of the instalments, increasing the Non Performing Loans (NPLs) of the banks.

With the insights from the research of Hyman Minsky (Minsky, 1993), we try to argue that the financial liberalization makes the economy more inclined to BC (banking Crisis) and CC (currency crisis). There is lack of empirical investigation studying the linkages financial openness and the financial system turmoil or fragility in the financial and economic system, thus study is an attempt to fill this gap.

2. Review of Literature

(Caprio & Klingebiel, 1997) the bank insolvencies are increasing due to systemic failure, weak institutional policies which drain the country's finances. (Aslı Demirgüç-Kunt & Detragiache, 1998) analyze the empirical relationship between financial liberalization and banking crises using the data from 1980-1995 for 53 nations. Their findings suggest that in a liberalized financial system the probability of the banking crises is more likely. But in a stable and a strong institutional environment the impact of liberalization on the vulnerability of banks is lesser. (G. Kaminsky, 1999) analyzed the link between the 76 currency crisis and 26 banking crisis for the period 1970 to 1995 taking into account 20 countries again focusing on the fragility of economy as an indicator of future crisis. But (Demirguc-Kunt & Detragiache, 2000) uses a multivariate logit technique to develop the model of banking crisis that can be used for monitoring the vulnerabilities in the banking sector. The results are lower in type I and type II error than the previous approaches of determinants of banking crisis. After investigating 90 banking crises, 202 currency crisis and 37 twin crisis (Glick & Hutchison, 2002), finds the more liberalized emerging economies are prone to twin crises and mainly concentrated on openness triggers excessive booms and busts cycles in financial markets leading to crisis. (Graciela L. Kaminsky & Reinhart, 1999) focuses on the issue of both banking and balance of payment crises, and states that with the sudden liberalization in

the financial sector leads to a shock to financial institutions as the capital flow increases, also the access to capital markets abroad fuels the boom period of the economy providing excessive financing options. (Graciela L Kaminsky & Schmukler, 2003; Graciela Laura Kaminsky & Schmukler, 2008; Ranciere, Tornell, & Westermann, 2006) supports the same school of thought with little difference in the findings. They show that financial openness in the short run causes larger investment cycles witnessing the development of lending boom which ends up in financial distress.

(Rivera-Batiz, 2001) constructed an endogenous growth model in which corruption is a vital part of the bureaucratic system to study the relationship between CAC and growth and concludes that if the level of corruption is high in the system the liberalization boost capital outflows. Similarly (Blackburn & Forgues-puccio, 2008) summarized their study stating that the effect of corruption is more adverse in an open economy than the closed one and in a weak bureaucratic system. The growth of an economy is far less in presence of corruption even if the capital account is made open and foreign flows is invited for investment. If the governance is weak the investors will lose the confidence in the long run finding it riskier and less profitable to invest in the nations with weak rule of law, (Elbir & Goaid, 2012) in their paper suggest that the lower corruption and financial liberalization are substitutes in promoting growth. There is clear evidence from literature that in the developing nations, if the institutional environment is weak, the financial liberalization has costs in terms of increased fragility in the economy and the financial markets. This may even spread from one nations to many other in the integrated and liberalized markets. It is really very important to develop the strong institutional environment before taking up full capital account liberalization. (Beck, 2008; Hellmann, Murdock, & Stiglitz, 2000) suggest that due to the regulatory and supervision failures, the financial liberalization and unfettered competition globally, has resulted in the systemic failures and undermine prudential bank behaviour. Similarly (Giannetti, 2007; Noy, 2004) in their research writes that one of the main culprits of volatility in the banking sector is financial liberalization due to the sudden loss of monopoly power of the domestic regulated bank as the uninformed international lend invest large amount funds at low cost.

3. Data and Methodology

The data set adopted for the purpose of study is for the years 1980-2005 and covers BRICS (Brazil, Russia, India, China and South Africa), ASEAN 5 [Indonesia, Korea, Malaysia,

Philippines and Thailand (those impacted by 1997-98 crisis)] also ALL representing the above mentioned 10 countries. These nations are selected as the sample as they are major emerging markets as per the definition from World Economic Outlook, IMF.

3.1 Banking Crises

The banking crises is adopted from the (Reinhart, 2010) based on a country to country analysis of various forms of economic and financial crises. The definition of the banking crises is adopted from the (Reinhart & Rogoff, 2008) in which they basically analyzed the stresses events in the banking sector. They take into account the two type of events: “ (1) bank runs that lead to the closure, merging, or takeover by the public sector of one or more financial institutions and (2) if there are no runs, the closure, merging, takeover, or large-scale government assistance of an important financial institution (or group of institutions), that marks the start of a string of similar outcomes for other financial institutions”. But researchers criticised this method as the closure of the banks can be too late after the actual problem with the bank. On the similar pattern (Laeven & Valencia, 2012) defined the banking crises dating process, the second criterion mentioned above is adopted is the first one is not met. One of the reasons for adopting the data base is the availability of data till 2010, so it helps to capture the 2007-09 period of financial turmoil.

We use the binary variable where 1 represents the year of banking crises (onset of the banking crises) and 0 otherwise. A total of 77 episodes were identified for all the nations included in the sample during the defined period (See the appendix A). Out of the 77 crises episodes 19 were recorded as the twin crises. The twin crises are the situation when the nations bear both the banking and the currency crises. The currency crisis is defined as a nominal depreciation of 25% or greater in the value of the currency, which is at least 10% greater than the depreciation in the previous year (Frankel & Rose, 1996) but (Milesi-ferretti & Razin, 1998) suggest that the previous year depreciation is below 40% and the value of depreciation should double that the preceding year..

3.2 Financial Liberalization

There are different categories of measurement of financial liberalization or upliftment of controls from the Capital account³:

- De Jure⁴

³ We are considering the financial liberalization as openness of the capital account.

- De Facto⁵
- Hybrid Measures⁶

For the purpose of the research we take into account only De Jure measures as they are directly related to the policies and regulatory aspect of the nation. Any change in this is related to the change in the policies of a nation in relation to financial liberalization. The de jure measure takes into account the scale based on the rigorousness and the degree of capital controls or in particular the binary variables from the AREAER recorded by IMF. These binary variables include the restrictive and non-restrictive regulations in relation to capital controls further diversifying in according to the resident and the non resident categories. In these studies the 1 indicates the liberalization or deregulation of control and 0 represents any kind of restriction imposed. The Annual report on Exchange Arrangements and Exchange Restrictions was a binary Measure until 1996 and then recorded with the sub Categories subsequently. For the liberalization (the openness of the Capital account) of the economy we adopted the data set from (Abiad, Detragiache, & Tressel, 2008) who measured the intensity of the openness of the financial sector by taking into account the different dimensions and scoring on the scale from 3 (fully liberalized) to 0 (not liberalized at all). It also take into account the policy reversals (but not measured in the paper). The first dimension is the credit controls and reserves requirement focusing on the credit ceilings, the next dimension is interest rate controls than is entry barriers followed by the state ownership. The next important dimensions are the restriction on the international capital flows, security market restrictions and the last is the prudential banking regulations (a greater score in the prudential regulation dimension would mean better regulation).

3.3 Bureaucracy, Lax Supervision and Economic Freedom

For the purpose of the study we try to evaluate the bureaucracy environment and the failure of enforcement of prudential regulation and economic freedom as a whole. Regulatory Efficiency takes into account the amount of government intervention into the Legal, Economic and Monetary aspects of a country. The freedom is based on three fundamental principles: empowerment of the individual, non-discrimination and open competition leading to the creation and maintenance of mutual sense of liberty for all. The Economic freedom

⁴ De Jure measures are the binary indicators developed for the controls on the capital account i.e. the cross border capital flows. (the 0 and 1 coding for the presence and the absence of the capital controls)

⁵ De Facto measures in the captures the real market situation taking into the account the integration of a nation into the international capital and money markets (Lane & Milesi-Ferretti, 2001).

⁶ Blend of both De Jure and De facto.

should provide an absolute right of property ownership, free movement of labour and capital, consumption of goods and services, economic decisions at the decentralized form. Various previous researches as mentioned earlier in the section of review of literature have reported in their studies the relationship between the ‘lax supervision’ and the failure of the financial liberalization. Therefore it becomes important to study the bureaucratic environment of the nation before commenting of the volatility in the banking and the financial sector. So, we used the Economic Freedom of the World Index⁷ developed by the FRASER Institute based on 42 quantitative and qualitative data points recorded separately under 5 different categories:

1. size of government: expenditures, taxes, and enterprises;
2. legal structure and security of property rights;
3. access to sound money;
4. freedom to trade internationally;
5. regulation of credit, labor, and business.

3.4 Macroeconomic Variables

To avoid the spurious regression problem we added various macroeconomic and financial variables that have been mentioned in the earlier studies. These are used as the control variables as they have a significant impact on the banking crises. We selected these variables from the list mentioned in the (Demirguc-Kunt & Detragiache, 2000; Barry Eichengreen & Arteta, 2000; Glick & Hutchison, 2002; Mehrez & Kaufmann, 2000). The various variables used in the regression were: GDP growth, Terms of Trade adjustment, Real Interest Rate, Real effective Exchange Rate, Inflation (Consumer price Index), Money and Quasi Money to Total Reserve Ratio, Bank Non-Performing Loans to Total Gross Loans, Domestic Credit to Private Sector Loans and Inflation GDP deflator (See appendix B for the definition of Variables and Sources)

3.5 Panel Data Probit Model Specification

As the dependent variable in the study i.e. the banking crises is binary in nature (representing 1 for the crises period and 0 otherwise) therefore we employ panel data probit model as given in equation (1). In the equation LHS is the dependent variable $BNK_{(i,t)}$ (banking crises) and we assume that crises will occur at time t for country i is a function of a vector of n variables

⁷ For the detail methodologies review the study: Gwartney, James, Hall, Joshua, and Lawson, Robert (2014). Economic Freedom of the World: An Accounting of the Literature. Contemporary Economic Policy. <http://onlinelibrary.wiley.com/doi/10.1111/coep.12010>.

denoted by $X(i,t)$, that includes control variables ($CTRL_{i,t}$) and the other independent variables (see appendix B). β is a vector of unknown coefficients and $F(\beta'X(i,t))$ is the cumulative probability distribution function. The probability function of the model can be written as:

$$\ln L = \sum_{t=1}^T \sum_{i=1}^n \left[BNK(i,t) \ln \{F(\beta'X(i,t))\} + (1 - BNK(i,t)) \ln \{1 - F(\beta'X(i,t))\} \right] \quad (1)$$

$$BNK_{i,t} = \beta_{i,t} + \lambda(LIB_{i,0}) + \psi \left[\sum_{i=1}^4 (\Delta LIB)_i \right] + \theta(ECFR_{i,t}) + \eta \left[\sum_{i=1}^4 (\Delta LIB)_i \right] * (ECFR_{i,t}) + \gamma(CTRL_{i,t}) + \xi_{i,t} \quad (2)$$

In the probit model the estimated coefficients capture the effect of change in an explanatory variable, where the sign of the coefficient implies the direction of the change, whereas the slope of the cumulative distribution function defines the magnitude of change. So the probit model is appropriate for the purpose of the as we want to analyze how does the financial liberalization reforms effects the occurrence or non occurrence of crises. So, we use panel containing yearly observations the defined set of indicators for the BRICS and A5 nations. The variables in the data are stationery and do not have endogeneity problem.

Equation 2: The banking crises variable $BNK_{i,t}$ is the likelihood function of crises in country i at time t at the intial level of liberalization, the $ECFR_{i,t}$ is the level of economic freedom and $CTRL_{i,t}$ is a matrix of control variables following (A Demirgüç-Kunt & Detragiache, 1998).

The Key Hypothesis: the purposed hypothesis is to test the effect of financial reform (Liberalization) variable on the probability of banking crises, conditional on economic freedom and macroeconomic variables.

4. Empirical Results for Financial Liberalization and the Banking Crises

The results are shown using the panel Probit model and analyze the relationship between the banking crisis and financial liberalization with the regulatory and macroeconomic control variables. We run the different regressions for the set of emerging markets i.e. A5 separate from the BRICS nations and a combined regression for both set of countries (ALL). In this paper we try to compare the two groups of nations, 1st being the A5 (those suffered with crises) being more liberalized and the 2nd group is BRICS which are less affected by crises (being less or partially liberalized). The third situation we club all the 10 nations and make an analysis. We have run the regression in three different sets:

4.1 Effect of financial reforms on banking crises (Table 1)

4.2 Effect of economic freedom on banking crises (Table 2)

4.3 Effect of financial reform and economic freedom on banking crises (Table3)

4.1 We estimate the equation using the Panel Probit model with the Random effects and report the results in table 1. We regress banking crises only on the financial reform variables and control variables. We report the regression with average marginal effects on means because they are more meaningful and accurate measures, as suggested in literature. Our findings are in line with those of the previous studies.

In countries(1) We analyze the relationship between the total score of financial reform variable and found that it is not significant, but when we segregate the Financial reform variables into different categories as mentioned in(Abiad et al., 2008) we found the results significant in few cases. In set countries(2) The Banking Entry barriers come out to be significant variable in all three groups. But when we compare the A5 and ALL nations set we find that Privatization increases the occurrence of the crises. In case of A5 international capital flows are also significant in increasing the probability of crises. In case of control variables the real GDP growth, Broad Money to reserves and domestic credit to private sector also predict the chances of crises.

Table 1 Effect of Financial Reforms on the Banking Crises

Independent Variables	COUNTRIES(1)			COUNTRIES(2)		
	BRICS	A5	ALL	BRICS	A5	ALL
Financial Liberalization:						
CRT CO				.4623 (.3805)	.0793 (.2302)	.0664 (.1643)
INT CO				-.5345 (.3641)	-.3538* (.2086)	-.2125 (.1443)
BAN EN				.8587** (.4007)	.5171* (.2751)	.5029*** (.2110)
PVT				-.4089 (.3680)	-.0807*** (.2414)	-.7008*** (.2224)
SUP CO				.0980 (.3734)	.5217 (.4196)	.2615 (.2140)
CAC CO				-.3550 (.3002)	-.4760* (.2667)	-.1759 (.1671)
SCR MA				.0468 (.3558)	.1141 (.4553)	-.0301 (.2285)
LIB (Financial Reform)	.0289 (.0297)	-.0396 (.0438)	.0019 (.0222)			
INDEPENDENT CONTROL						
GDP GROW	.0645 (.0457)	-.1978*** (.0462)	-.0508** (-.0244)	.0569 (.0537)	-.1597*** (.0502)	-.4012 (.0289)
GDP/CAP	-.0000 (.0000)	-.0004 (.0003)	-.0006* (-.0003)	.0001 (.0001)	-.0001** (.0000)	-.0001** (.0000)
TOT	.0000	.0000	.0000	-.0000**	.0000	.0000

	(.0000)	(.0000)	(.0000)	(.0000)	(.0000)	(.0000)
RE INT	.0046	-.0035	-.0081	.0041	.0048	-.0044
	(.0087)	(.0284)	(.0075)	(.0118)	(.0352)	(.0088)
REER	-.0009	.0008	-.0030	.0020	.0090**	-.0010
	(.0029)	(.0027)	(.0003)	(.0036)	(.0044)	(.0038)
INF CP	.0008**	-.0616**	.0002	.0008	-.0492	.0001
	(.0003)	(.0283)	(.0003)	(.0005)	(.0340)	(.0003)
M2/TR	-.0124	.2340***	-.0029	.0083	.3359***	.0259
	(.0228)	(.0684)	(.0193)	(.0289)	(.0867)	(.0217)
CRD PVT/GDP	.0044	.0079*	.0051	-.0122*	.0124*	.0030
	(.0052)	(.0045)	(.0036)	(.0070)	(.0065)	(.0049)

***, **, * indicates rejection of null(coefficient=0) at 1%, 5%, 10% significance levels, respectively.

4.2 In table 2 we regress the dependent variable banking crises on the economic freedom index. In the first regression (countries1) we found that in case of BRICS nation the Economic freedom comes out to be significant where as in A5 and ALL it is not. But when we run the second regression (countries 2) we found that legal system and property rights, Sound money, freedom to trade internationally and regulatory system are significant in BRICS nations at 1% level. The regulation variable is significant in all three sets in predicting crises.

ENDOGENEITY:

Though we followed various previous studies and examine the impact of financial liberalization and economic freedom on banking crises but the regression may suffer from endogeneity problem. This problem may occur because due to crises certain time the regulators may reemploy a policy or reform or reverse the liberalization. We dropped all the observations after the initial year of the crises and end of the same crises period. Than we run the regressions again and concluded that there is no difference in the previous results.

Table 2 Effect of Economic Freedom on the Banking Crises

Independent Variables	COUNTRIES(1)			COUNTRIES(2)		
	BRICS	A5	ALL	BRICS	A5	ALL
SOG				.1886	.1490	.3149**
				(.3367)	.2691	(.1445)
LSPR				-1.371***	-.2684	-.1624
				(.3872)	(.1950)	(.1425)
SM				-.8196***	-.1072	-.1327*
				(.2638)	(.1313)	(.0729)
FTI				.5474***	.3077	.1724
				(.2085)	(.2794)	(.1148)
REG				-.7656***	-.6516**	-.4684***
				.2701	(.3009)	(.1769)
Economic Freedom (total)	-.3218*	-.3099	-.1131			
	(.1845)	(.2863)	(.1408)			
INDEPENDENT CONTROL						
GDP GROW	.0437	-.1987***	-.0506**	.1081	-.1951***	-.0478*

	(.0450)	(.0467)	(.0244)	(.0712)	(.0506)	(.0268)
GDP/CAP	-.0001	-.0001	-.0001*	-.0006**	-.0000	-.0001
	(.0001)	(.0000)	(.0001)	(.0002)	(.0000)	(.0001)
TOT	.0000	.0000	.0000	.0000***	.0000	.0001
	(.0000)	(.0000)	(.0000)	(.0000)	(.0000)	(.0001)
RE INT	.0019	-.0003	-.0085	-.0089	-.0047	-.0079
	(.0087)	(.0270)	(.0076)	(.0125)	(.0319)	(.0089)
REER	-.0033	-.0005	-.0032	-.0052	.0005	-.0034
	(.0030)	(.0027)	(.0026)	(.0069)	(.0326)	(.0036)
INF CP	.0006	-.0589**	.0002	.0004	-.0511	-.0001
	(0003)	(.0271)	(.0003)	(.0004)	(.0326)	(.0001)
M2/TR	-.0175	.2325***	-.0076	.0046	.2656***	.0046
	(.0221)	(.0683)	(.0196)	(.0319)	(.0765)	(.0215)
CRD PVT/GDP	-.0001	.0102**	.0067*	-.0064	.0193***	.0103**
	(.0057)	(.0052)	(.0038)	(.0085)	(.0067)	.0048

***, **, * indicates rejection of null(coefficient=0) at 1%, 5%, 10% significance levels, respectively.

4.3 In table 3 the third set of regression we interpreted that only in the case of BRICS nations(Countries1) the total financial liberalization and economic freedom variable are the predictors of crises. But the results for A5 and ALL the nations doesn't support the hypothesis, the signs of the variables are as per the previous studies that if the liberalization is more and the economic freedom is less the chances of crises increases. In the case of countries2 regression we regress banking crises on individually on all the components of financial reform variable and the economic freedom variable. In case of BRICS nation the Securities market openness comes out to be negatively significant, this is because these nations have less or partially liberalized capital accounts. The economic freedom components also are on same line as more the freedom and repressed financial system more is the chances of crises. The sound money, LSPR, FTI and regulation are highly significant at 1% level.

In the group of A5 and ALL nations the Interest rate control, privatization in financial reform and LSPR are common predictors of crises. But in ALL banking entry barriers in financial liberalization and system of government, sound money and regulation are predictor of crises.

In the countries3 for ALL the nations we run a regression without including the control variables the similar predictors of crises. These significant variables show the occurrence of the crises in different situations. More liberalized and less developed nations increases the chances of occurrence of the crises.

Table 3 Effect of Financial Reforms and Economic Freedom on the Banking Crises.

Independent Variables	COUNTRIES(1)			COUNTRIES(2)			COUNTRIES(3)
Financial Liberalization:	BRICS##	A5#	ALL	BRICS##	A5#	ALL	ALL###
LIB (Financial Reform)	.2490***	-.0160	-.2601				
	(.0781)	(.0563)	.2100				
CRT CO				1.3056	.4819	.1774	.1416
				(.0853)	(.3283)	(.1837)	(.1607)
INT CO				.1368	-.7606***	-.4507***	-.4005**
				(.7850)	(.2833)	(.1841)	(.1667)
BAN EN				.9190	.1424	.5899**	.6922***

				(.6158)	(.4006)	(.2612)	(.2445)
PVT				.6219	-.7998***	-.9306***	-.9638***
				(.1550)	(.2864)	(.2450)	(.2124)
SUP CO				.2151	.6727	.3433	.4211*
				(.0294)	(.5782)	(.2794)	(.2427)
CAC CO				-.3769	-.2272	.1110	-.0143
				(.6075)	(.3302)	(.2012)	(.1827)
SCR MA				-1.7408**	.6514	-.3449	-.3075
				(.0643)	(.5535)	(.2794)	.2524
Economic Freedom (total)	-1.4699***	-.2442	.0319				
	(.4051)	(.3677)	(.0329)				
SOG				.1018	.3462	.6098***	.6544***
				(.6681)	(.4421)	(.2196)	(.2237)
LSPR				-1.5397***	-.5042**	-.3765**	-.3987**
				(.5486)	(.2546)	(.1790)	(.1678)
SM				-2.0377***	-.2473	-.2174***	-.1949**
				(.0849)	(.1738)	(.0908)	(.0818)
FTI				.9601**	.3860	.0915	-.0498
				(.5082)	(.0403)	(.1668)	(.1414)
REG				-1.6690***	-.5607	-.6157***	-.5649***
				(.0312)	(.4137)	(.2116)	(.1938)

***, **, * indicates rejection of null(coefficient=0) at 1%, 5%, 10% significance levels, respectively. [#For A5 nations in the regression all the Economic control variables (except TOT, RE IN, Inflation) were significant at 1% level of significance. ## For BRICS nations GDP/ Cap, TOT, RE INT, REER are significant at 5% levels].### is the regression equation where no control macroeconomic variable was used]

5. Conclusions

In this paper, we examine the effect financial liberalization on the likelihood of banking crises. For the purpose we had regressed the different dimensions of financial reforms and the economic freedom on the odds of banking crises. We find that if financial liberalization is undertaken in the weak institutional environment and lack of economic freedom and financial instability the likelihood of banking crises augment. Our results also hold after controlling for a possible endogeneity problem. We also found that stronger GDP Growth rate reduces the probability of banking crises whereas the credit to private sector increases the same. The paper supports for the gradual approach towards the liberalization.

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Appendix A Banking Crises Episodes

Brazil	1994-1997
China	1992-99
India	1993-98
Indonesia	1997-2002
Korea	1997-2002
Malaysia	1997-2001
Philippines	1997-2001
Russia	1995,1998, 2008-09
South Africa	-
Thailand	1996-2001
United States	2007-2010

Appendix B Data Sources for Variables Included in the Probit Regression

Variable Name	Description	Expected Sign	Source
Dependent Variable			
BANK CR	Banking Crisis		(Reinhart & Rogoff, 2008) and (Laeven & Valencia, 2013)
Independent Variables			
LIB	Liberalization(Financial Reform)	+/-	(Abiad et al., 2008)
CRT CO	Credit Controls	+/-	(Abiad et al., 2008)
INT CO	Interest Rate Controls	+/-	(Abiad et al., 2008)
BAN EN	Banking Entry	+/-	(Abiad et al., 2008)
PVT	Privatization	+/-	(Abiad et al., 2008)
SUP CO	Supervisory Control	+/-	(Abiad et al., 2008)
CAC CO	Capital Controls	+/-	(Abiad et al., 2008)
SCR MA	Securities Market	+/-	(Abiad et al., 2008)
SOG	Size of Government	+/-	(James et al., 2014)
LSPR	Legal Structure and security of property Rights	+/-	(James et al., 2014)
SM	Access to Sound Money	+/-	(James et al., 2014)
FTI	Freedom to trade internationally	+/-	(James et al., 2014)
REG	Regulation of Credit and Business	+/-	(James et al., 2014)
INDEPENDENT CONTROL VARIABLES			
(A Demirgüç-Kunt & Detragiache, 1998)			
GDP GROW	Rate of Growth of Real GDP	-	World economic outlook, IFS*, IMF**
GDP/CAP	GDP per capita growth	-	World economic outlook, IFS, IMF
TOT	Change in Terms of Trade	+/-	World economic outlook,

			IFS, IMF
RE INT	Nominal Interest rate Adjusted for Inflation	+	World economic outlook, IFS, IMF
REER	Inflation Adjusted Exchange Rate	+	World economic outlook, IFS, IMF
INF CP	Inflation, GDP deflator (annual %)/NY.GDP.DEFL.KD.ZG	+	World economic outlook, IFS, IMF
M2/TR	Money and quasi money (M2) to total reserves ratio	+/-	World economic outlook, IFS, IMF
NPA/GL	Bank nonperforming loans to total gross loans (%)/FB.AST.NPER.ZS	+	World Bank, BIS***
CRD PVT/GDP	Domestic credit to private sector (% of GDP)/FS.AST.PRVT.GD.ZS	+/-	World Bank, BIS

*IFS: International Financial Statistics, **IMF: International Monetary Fund, ***BIS: Bank for International Settlements